

SUPPLEMENT ARTICLE

Wearable devices for sudden unexpected death in epilepsy prevention

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Summary

Sudden unexpected death in epilepsy (SUDEP) is most often associated with the occurrence of generalized tonic–clonic seizures (GTCS), a seizure type that can now be detected with high sensitivity and specificity by wearable or bed devices. The recent development in such devices and their performance offer multiple opportunities to tackle SUDEP and its prevention. Reliable GTCS detection might help physicians optimize antiepileptic treatment, which could in turn reduce the risk of SUDEP. GTCS-triggered alarms can lead to immediate intervention by caregivers that are also likely to decrease the odd of SUDEP. The biosignals used to detect GTCS might provide novel SUDEP biomarkers, in particular, by informing on several important characteristics of the ictal and postictal periods (type of GTCS, duration of tonic phase, rotation in the prone position, presence and duration of postictal immobility and bradycardia, rise in electrodermal activity). Other biosensors not yet used for detecting GTCS might provide complementary information, such as the presence and intensity of ictal/postictal hypoxemia. The above biomarkers, if strongly predictive, could help identify patients at very high risk of SUDEP, enabling better assessment of individual risk, as well as selection of appropriate patients for clinical studies aiming at preventing SUDEP. The same biosignals could also be used as ancillary biomarkers to test the impact of various interventions before moving to highly challenging randomized controlled trials with SUDEP as a primary outcome.

KEYWORDS

epilepsy, generalized tonic–clonic seizure, SUDEP, wearable devices

1 | INTRODUCTION

Although significant progress has been made in the field of sudden unexpected death in epilepsy (SUDEP) over the past decade, in particular regarding its underlying pathophysiology,^{1,2} we are still in urgent need of appropriate biomarkers to more precisely delineate individual risk of SUDEP and preventive interventions to reduce that risk.

It is our view that wearable and connected devices offer a high potential to tackle these issues, and more specifically to provide biomarkers that will help us to further understand the mechanisms of SUDEP and identify patients at high risk.

This would allow the testing of interventions that might prevent SUDEP, either through optimization of antiepileptic treatment, detection of ongoing seizure and/or cardiorespiratory arrest, or triggering of resuscitation procedures.

2 | IDENTIFYING BIOMARKERS AND MECHANISMS OF SUDEP

Our current understanding of the primary mechanisms of SUDEP suggests that the occurrence of generalized tonic–clonic seizure (GTCS) plays a critical role in the majority

of cases, with most witnessed SUDEP immediately following a GTCS.¹ In addition, a GTCS frequency ≥ 3 /year is associated with an odds ratio of 15.6 (95% confidence interval [CI] 10.1-24.2) as compared to no GTCS.³ Still, the annual risk of patients with ≥ 3 GTCS/year, extrapolated from the combination of epidemiology and case-control studies, is estimated between 0.5 and 1.8/100 patient-years (PYs),⁴ which means that only a minority of patients with frequent GTCS will die of SUDEP, and that these patients usually have a large number of GTCS before a fatal seizure. Several patient-related factors might contribute to the lethality of GTCS, including electroclinical characteristics, time of occurrence and propensity to result in a prone position, patient's medications, and innate or acquired traits influencing the impact of GTCS on brainstem cardiorespiratory centers. Some of these risk factors can be readily identified, such as male gender, intellectual disability, and the occurrence of nocturnal seizures,⁵ or only require once-in-a-lifetime investigations, such as an extratemporal epileptogenic zone.⁵ Others might require repeated assessments due to intraindividual variability over time, such as the presence and duration of postictal generalized electroencephalography (EEG) suppression (PGES).^{6,7} However, performing repeated video-EEG to characterize the severity of GTCS in a given patient is not a realistic option.

Conversely, wearables and connected devices that would provide a surrogate marker of PGES could be used long-term in ambulatory patients to provide information on the presence of this risk factor.⁸ Several such surrogate markers can be envisioned based on their known association with PGES. Indeed, the latter was found associated with the duration of postictal immobility^{9,10} and postictal coma,¹¹ the duration of the tonic phase,¹¹⁻¹⁴ the type of GTCS,¹² and the ictal/postictal changes in electrodermal activity (EDA).^{15,16} A correlation was also reported between PGES and the duration of the clonic phase in one study,¹³ but not in others.^{11,12,14} Most of the preceding PGES-associated variables can be readily measured by existing sensors incorporated into commercially available GTCS-detection devices, such as accelerometers¹⁷⁻¹⁹ and electromyography (EMG)²⁰ for postictal immobility and duration of tonic or clonic phases, and skin conductance sensors for EDA.^{21,22} The distinction of the various types of GTCS with such devices also appears feasible, although this remains to be demonstrated.

Because we still ignore the causal relationship between PGES and the above-listed variables, it might well be that the latter proves more predictive than PGES itself. Such information, gathered from wearable devices, would provide novel insights into the sequence of events resulting in SUDEP. For instance, an unusual fast rising and large postictal electrodermal response was observed in a patient who died of SUDEP while wearing an EDA-based GTCS-detecting wrist device.²³

Key Points

- Detection of generalized tonic-clonic seizures (GTCS) by wearable devices might allow for optimization of the chronic and acute management of GTCS, which could in turn result in reducing the risk of GTCS
- Wearable devices offer unique opportunities to develop novel biomarkers to more precisely assess the individual risk of SUDEP
- Wearable devices shall facilitate the development of clinical trials for testing interventions to prevent SUDEP

Other potential risk factors of SUDEP that deserve to be investigated include postictal cardiorespiratory changes, in particular postictal bradycardia/asystole and hypoxemia,²⁴ and whether the patient is in a prone position,²⁵ all of which are easily captured by existing wrist-based or bed-attached sensors.²⁶⁻²⁸ Connected T-shirts in development, with various thoracic sensors, can also capture electrocardiography (ECG) features of interest, such as QT length, heart rate variability, and T-wave alternans, but maybe at the price of some discomfort. Some of these features might also be extracted from biosignals measured at the devices, though raising both signal quality and battery-life issues.²⁹ Measurement of blood pressure and detection of stertor, which could be selectively triggered following a GTCS detection, could also be of value.³⁰

Most importantly, the capacity to continuously monitor relevant biomarkers over very long periods of time (months, years) using wearable devices, offers the possibility of investigating how SUDEP risk factors might evolve in a given patient until SUDEP occurs. This dynamic, and intra-individual relative variation of variables of interest could prove a better predictor of SUDEP than punctual and absolute biomarker values.

Finally, biosignals other than those currently detectable with existing wearable devices, including biochemical biomarkers, are likely to be developed in the future and could further improve our ability to establish the individual risk of SUDEP.

3 | IDENTIFYING PATIENTS AT VERY HIGH RISK OF SUDEP

The average risk of SUDEP in patients with epilepsy is estimated at 1.2/1000 PY (95% CI 0.64-2.32) in adults, and 0.22/1000 PY (95% CI 0.16-0.31) in children.⁵ However, these average figures pool together populations where risk

of SUDEP varies by almost a factor of 100,³¹ with the greatest risk reported in patients awaiting or having failed epilepsy surgery (9.3/1000 PY),³² and in those with refractory epilepsy allocated to placebo in add-on randomized controlled trials (6.9/1000 PY).³³ As discussed previously, extrapolation from case-control studies suggests that patients with more than 3 GTCS per year might have a maximal risk of SUDEP of 18/1000 PY.⁴ Although these latter figures are severe, inasmuch as many patients will endure such conditions over decades with a lifelong cumulative risk exceeding 10%,³⁴ they probably reflect a heterogeneous situation with subpopulations at higher risk than others. Indeed, case-control studies that demonstrated the major role of GTCS in promoting SUDEP did not distinguish between the different types of GTCS.³ Furthermore, and as already discussed, the risk of SUDEP is also likely to vary over time in a given individual.

Based on the preceding considerations, one might expect some patients to endure an annual risk of SUDEP that would exceed 2% and may be up to 5%. Identifying such patients is important, so as to advise them appropriately according to the currently known safety measures, but also represents a mandatory step toward clinical studies of preventive interventions. Indeed, as illustrated in Table 1, the sample size required to perform such a trial, based on a 0.4% yearly rate of SUDEP (average SUDEP risk in refractory epilepsy), a 50% or even 80% risk reduction with the tested intervention, and a 1-year follow-up, appears unrealistic, both from a financial and recruitment point of view. Indeed, patients with this level of risk are unlikely to be interested in such a study, let alone the fact that the most promising interventions might be implanted devices.⁴ Conversely, patients at very high risk (eg, $\geq 5\%$) are more likely to be willing to test potentially preventive interventions, and would offer a manageable sample size.

The potential for wearable and connected devices to provide highly predictive biomarkers or a combination of biomarkers, offers some hope for reaching these objectives. In addition, biomarkers that would be both highly predictive and modifiable could be used as surrogate markers of SUDEP to test the impact of preventive

TABLE 2 Theoretical example of an interventional study against SUDEP with an ancillary biomarker as primary endpoint

Strategy	Illustrative example
Select a biomarker predictive of SUDEP and reliably captured by a wearable device	Non-EEG correlate of PGES >20 s (eg, combination of postictal immobility and EDA above a specific threshold)
Select population at high risk of SUDEP	Patients with ≥ 1 GTCS/month and 3 consecutive GTCS with biomarker value above threshold
Hypothesize that intervention modifies biomarker	Biomarker above threshold is reduced by 50% in intervention group vs 5% in control group
Record one GTCS per patient during the intervention study	34 patients need to be recruited to test the above hypothesis with bilateral test, $\alpha = 0.05$, $1 - \beta = 0.9$

interventions in much smaller populations and over much shorter periods of time. As illustrated in Table 2, let's assume that a combined measurement of post-GTCS immobility and EDA, using a wrist-based device, allows for the prediction of PGES >20 seconds with a very high sensitivity and specificity. One could then use such measurements to test the impact of any given intervention with a much smaller sample size (eg, about 30 patients) and study duration (until one GTCS occurs in each recruited patient) than those needed if SUDEP was a primary endpoint (see Table 2).

This approach would thus allow the selection of the most promising prevention strategy that should further be tested in the much more challenging and expensive randomized controlled trials discussed earlier.

The modifiable nature of the biomarker is essential to consider, although not necessarily easy to determine, inasmuch as it depends on the type of intervention used. For instance, antiepileptic treatment might reduce the severity of GTCS by interfering with the propagation and sustainability of the ictal discharge. GTCS characteristics such as the type of GTCS and the duration of the tonic phase might then be considered and used as modifiable biomarkers. Conversely, other drugs such as selective serotonin reuptake inhibitors, might act primarily downstream of the ictal phase by reducing the postictal phenomenon thought to participate in the emergence of PGES and brainstem dysfunction, regardless of the duration of GTCS or its tonic phase.^{35,36} For testing such interventions, modifiable biomarkers of interest would rather be those reflecting brainstem dysfunction (postictal immobility, hypoxemia, or bradycardia). Finally, some interventions, such as implanted medical devices, might selectively operate when a postictal cardiorespiratory arrest is detected, without the possibility of testing the efficacy of such devices on an upstream ancillary biomarker.

TABLE 1 Number of patients to be randomized to demonstrate the significant impact of an intervention against SUDEP during a 2-year follow-up study (unilateral test, $\alpha = 0.05$, $1 - \beta = 0.9$)

Risk of SUDEP in control group	0.4% / year	2% / year	5% / year
If intervention reduces risk by 50%	N = 12404	N = 2422	N = 924
If intervention reduces risk by 80%	N = 3488	N = 686	N = 264

4 | OPTIMIZING EPILEPSY CARE

Most SUDEP are triggered directly by a seizure; thus it is likely that optimizing antiepileptic treatment should translate into a reduced risk of SUDEP. This is strongly suggested by our meta-analysis of all randomized controlled trials of antiepileptic drugs (AEDs) performed in patients with drug-resistant epilepsy, which showed a 7-fold lower rate of SUDEP in patients allocated to an add-on AED as compared to those allocated to an add-on placebo.³³ Some data also suggest that successful epilepsy surgery might prevent SUDEP,^{37,38} although a formal demonstration of this impact is still missing.³⁹

Wearable devices might help the optimization of antiepileptic treatment in several ways relevant to the risk of SUDEP. First, it might help provide physicians with a more precise account of seizure frequency and its evolution following changes in AED regimen. Indeed, it is well known that patients' own report of their seizure frequency is often unreliable, with up to 55% of nocturnal GTCS going unreported.⁴⁰ Furthermore, AEDs might have a favorable or unfavorable impact on various aspects of GTCS severity as well as on the risk of SUDEP, regardless of their influence on seizure frequency. For instance, a few reports have suggested that lamotrigine and carbamazepine could be associated with a higher risk of SUDEP,^{41–43} although these findings were not confirmed in a meta-analysis.³ Similar possibilities arose with vagus nerve stimulation, which is known to reduce GTCS frequency and the duration of postictal confusion,^{44–46} but might also aggravate post-GTCS respiratory dysfunction through the same mechanisms by which it promotes sleep apnea syndrome in some patients.^{47–49}

Therefore, monitoring the impact of antiepileptic treatment on biomarkers of GTCS severity using wearable devices might help the adjustment of such treatments with a view to reduce the risk of SUDEP.

Wearable devices are usually coupled with a smartphone app,²⁸ hence offering opportunities to promote patient education regarding adherence to treatment, avoidance of seizure triggering factors (lack of sleep, alcohol, medications lowering seizure threshold, abrupt AED changes), and appropriate reaction to seizure clusters (rescue medication), missed medication (redosing), or to any other situations that could lower AED levels (gastrointestinal disorders, pregnancy, prescription of other drugs such as oral contraceptive in patients treated with lamotrigine).⁵⁰

5 | WARNING OF AN ONGOING SEIZURE

Several commercially available wearable devices are providing reliable detection of GTCS with an associated alarm

that can prompt immediate intervention from surrounding relatives or caregivers. These devices and their effectiveness are discussed in detail in other articles from this supplement.^{12,51,52}

Some evidence suggests that nocturnal supervision reduces the risk of SUDEP,^{42,53} indirectly supporting the view that timely postictal intervention by a caregiver could counteract some of the mechanisms at stake in SUDEP, such as the prone position. Other indirect arguments pointing to the same conclusion have been derived from epilepsy monitoring unit (EMU) data.¹ Mortality in Epilepsy Monitoring Unit Study (MORTEMUS).¹ Indeed, although all recorded SUDEP occurred in patients who did not benefit from any intervention within the 13 minutes following seizure termination, all near-SUDEP benefited from a cardiorespiratory resuscitation procedure within 3 minutes postictally (and most within 1 minute).¹ Furthermore, periictal nurse intervention in the EMU was found associated with a reduced risk of PGES.^{9,13} Of interest, smartphone technology enables not only the sending of a GTCS-triggered text message to predefined caregivers, but also offers the opportunity to send such messages to any other smartphone located in a given perimeter, which could promote life-saving interventions from GTCS witnesses outside the patient's home.²⁶

GTCS-triggered alarms might also enable early periictal oxygen administration. The latter was also found to reduce PGES in some studies,^{9,12} although its impact remains debated.¹³ In a mouse model of seizure-induced SUDEP, O₂ therapy proved extremely efficacious to prevent death.⁵⁴

6 | DETECTING IMMINENT OR ONGOING CARDIORESPIRATORY ARREST

Sensors available in wearable devices have the capacity to detect cardiac and respiratory distress, whether or not related to a seizure. Within the framework of a GTCS, this might allow specific alarms to be sent that would inform caregivers of the unusual severity and life-threatening potential of the seizure, prompting a more appropriate response, including cardiopulmonary resuscitation (CPR) when necessary. Indeed, caregivers who regularly assist patients with GTCS and are used to waiting patiently for the slow postictal recovery, might not necessarily detect the occurrence of an unusual cardiorespiratory dysfunction until it reaches a very late stage where CPR might be ineffective.¹ In most EMUs, nurses are trained to continuously check heart rate and SpO₂ following GTCS. Wearable devices equipped with appropriate sensors and alarms would allow caregivers to perform similar monitoring at home.

Detecting cardiorespiratory distress might also be useful outside seizures in patients with epilepsy at risk of

developing interictal malignant arrhythmias or respiratory arrest. Indeed, some patients with a genetic epilepsy resulting from mutation of voltage-gated sodium or potassium channel might be at risk of non-seizure-related sudden cardiac death.^{55–57} A few patients with epilepsy have also been shown to have a terminal subacute respiratory arrest of unknown origin without a preceding seizure.⁵⁸

Eventually, closed-loop medical devices will be developed for patients at very high risk of SUDEP, although the exact intervention to be triggered remains a matter of debate.⁵⁹ Cardiac pacing represents an easy option, but which might not be able to counteract the primary respiratory distress observed in most recorded SUDEP.¹ Stimulating respiratory muscles would represent another, nonexclusive, option. Although wearable devices could potentially trigger such interventions, it seems easier to rely on the more robust cardiorespiratory signal that could be recorded by the implantable device used for resuscitation.

7 | CONCLUSION

Overall, although still in their early stages of development, wearable devices offer multiple opportunities to prevent SUDEP. As of today, it can promote a timely response from caregivers following a GTCS. In the near future, it should allow the identification of new SUDEP biomarkers that can be measured chronically in ambulatory patients, with the view to (1) better understand the mechanisms of SUDEP, (2) identify patients at very high risk of SUDEP, and (3) test the impact of potential preventive treatments on modifiable biomarkers of SUDEP. Eventually, this progress will enable the development of SUDEP prevention trials. Wearable devices are likely to play a major role in the achievement of these trials as well as in the implementation of their findings, if positive, in clinical practice.

DISCLOSURE

None of the authors has any conflict of interest to disclose. We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

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